

Dynamic Optimization

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Introduction This part of the course studies how to solve *dynamic* models. Although such problems are particularly relevant in macroeconomics, they are also common in other fields. There are two main approaches to solving dynamic optimization problems — sequential and recursive ones. Within each approach, there are methods to solve (i) deterministic and stochastic problems in (ii) discrete and continuous time. These methods can be used to (a) solve the model in closed form, (b) obtain numerical solution, and (c) prove analytically some properties of the optimal solution. The choice between the approaches depends on the particular application and the goal of the analysis.

	discrete time	continuous time
sequential approach	Lagrangian	Hamiltonian
recursive approach	Bellman equation	HJB equation

This part of the course is closer to the cookbook — the main focus is on the mathematical tools that can be used to solve dynamic problems, the proofs are mostly heuristic, and simple macroeconomic models are used as examples.

Savings problem Consider a simple savings problem of a household living for T periods:

$$\begin{aligned} \max_{\{C_t, B_{t+1}\}} \quad & \sum_{t=0}^T \beta^t u(C_t) \\ \text{s.t.} \quad & B_{t+1} = RB_t + Y_t - C_t \\ & B_{T+1} \geq 0, \end{aligned} \tag{1}$$

where C_t and B_{t+1} are consumption and savings in period t , $R \equiv 1 + r$ denotes the gross returns on assets, and Y_t is an exogenous income. Equation (1) is the *flow budget constraint* of the household, while equation (2) restricts the agent to pay back any debt before her death. We make the standard assumptions about preferences:

1. discount factor $\beta \in [0, 1)$,

2. utility function $u(\cdot)$ is increasing and concave,
3. $\lim_{C \rightarrow 0} u'(C) = \infty$.

Optimality conditions Perhaps the simplest way to approach this problem is to apply the same optimization methods that are used to solve the static problem. The dynamic problem is indeed not that different from a static one given that consumption in different periods $C_0, C_1, \dots, C_T$ can be interpreted as consumption of different goods. To this end, write down the Lagrangian

$$\mathcal{L} = \sum_{t=0}^T \beta^t u(C_t) + \sum_{t=0}^T \lambda_t (RB_t + Y_t - C_t - B_{t+1}) + \mu B_{T+1},$$

where $\lambda_t, \mu \geq 0$, and take the first-order conditions (FOC):

$$\begin{aligned} \text{wrt } C_t &: \beta^t u'(C_t) = \lambda_t, \\ \text{wrt } B_{t+1} &: \lambda_t = R\lambda_{t+1}, \quad t < T, \\ \text{wrt } B_{T+1} &: \lambda_T - \mu = 0. \end{aligned}$$

As usual, the Lagrange multiplier λ_t reflects the shadow price of having an additional unit of good in period t and is equal to the margin utility of consumption in that period. The economic interpretation of the Lagrangian is that there are markets, in which the agent can exchange a unit of period- t good for λ_t units of utility or vice versa (think of this as warm-glow from charity or punishment for stealing). There is also a punishment (in units of utility) for leaving an unpaid debt when dying. In this sense, the optimization problem is fully static: all decisions are made in period zero and it only remains to execute the signed contracts in the following periods. This interpretation will prove useful below when we discuss the continuous-time version of the model. Combining the first two equations for $t < T$, we get the intertemporal optimality conditions, which are often called the **Euler equations (EE)**:

$$u'(C_t) = \beta R u'(C_{t+1}), \tag{3}$$

for $\forall t < T$. Several things to notice about this equation:

1. What is the mathematical intuition for the EE? The trajectory $\{C_t, B_{t+1}\}$ is optimal if any small achievable deviation from it has zero first-order effect on utility. E.g. consider a deviation in period t returning back to the trajectory in period $t + 2$, i.e. increase C_t by Δ , which lowers savings B_{t+1} by Δ and the next-period wealth by RB_{t+1} . To get back to the optimal trajectory, B_{t+2} needs to be unchanged and hence, from (1), we need to decrease C_{t+1} by RB_{t+1} . The net effect on welfare of this perturbation is

equal $dU = \beta^t u'(C_t) dC_t + \beta^{t+1} u'(C_{t+1}) dC_{t+1} = \beta^t u'(C_t) \Delta - \beta^{t+1} u'(C_{t+1}) R \Delta$. Thus, the trajectory is optimal and the deviation results in zero first-order effects if the Euler equation (3) is satisfied.

2. What is the economic intuition for the EE? One can rewrite it as standard $MRS = MRT$ condition: $\beta \frac{u'(C_{t+1})}{u'(C_t)} = \frac{1}{R}$. More importantly, it shows two saving motives: (i) consumption smoothing, (ii) intertemporal substitution βR – in particular, consumption increases in time when $\beta R > 1$.
3. Notice this is only one possible deviation and one optimality condition. One could instead focus on a two-period deviation and obtain $u'(C_t) = (\beta R)^2 u'(C_{t+2})$. Things only get more complicated when there are several choice variables, e.g. consumption and labor. The reason most of the literature focuses on equation (3) is because of its simplicity and intuitiveness.

The last period requires special treatment because of the complementary slackness condition

$$\mu B_{T+1} = 0,$$

i.e. either (i) constraint (2) binds and $B_{T+1} = 0$ or (ii) constraint (2) does not bind and $\mu = 0$. Combining this expression with the first-order conditions, we get the **transversality condition (TVC)**

$$\beta^T u'(C_T) B_{T+1} = 0, \tag{4}$$

which implies that the optimal bequest is $B_{T+1} = 0$. The economic intuition for the TVC is straightforward: either consumption should have zero marginal utility in the last period (which cannot be true under the assumed preferences), or it is optimal to leave zero savings and consume all funds before dying.

In sum, the optimal path $\{C_t, B_{t+1}\}$ with $2 \times (T + 1)$ unknowns is determined by $T + 1$ budget constraints (1), T Euler equations (3) and one transversality condition (4).

Example To solve the model in closed form, assume, for example, a constant relative risk aversion (CRRA) utility $u(C) = \frac{C^{1-\sigma}}{1-\sigma}$ and follow these steps:

1. Iterate forward the budget constraint

$$B_0 = R^{-1}(C_0 - Y_0 + B_1) = R^{-1}(C_0 - Y_0) + R^{-2}(C_1 - Y_1 + B_2) = \dots$$

to obtain

$$RB_0 = \sum_{t=0}^T R^{-t}(C_t - Y_t) + R^{-T} B_{T+1}.$$

2. Impose the terminal condition $B_{T+1} = 0$ to obtain the *intertemporal budget constraint*:

$$\sum_{t=0}^T R^{-t} C_t = RB_0 + \sum_{t=0}^T R^{-t} Y_t.$$

3. Express the Euler equation as

$$C_{t+1} = (\beta R)^{\frac{1}{\sigma}} C_t$$

and substitute into the budget constraint to solve for

$$C_0 = \frac{RB_0 + \sum_{t=0}^T R^{-t} Y_t}{\sum_{t=0}^T R^{-t} (\beta R)^{\frac{t}{\sigma}}}.$$

Infinite horizon So far, we assumed a finite T . However, it is often more convenient to focus on infinite horizon. This approach can also provide an accurate description of the real world if individual people care about the wellbeing of their descendants. Although the formal proofs are more complicated for infinite horizon, most results can be derived heuristically by simply taking the limit $T \rightarrow \infty$ in the expressions above. In particular, the household problem remains almost unchanged, except for constraint (2):

$$\max_{\{C_t, B_{t+1}\}} \sum_{t=0}^{\infty} \beta^t u(C_t)$$

$$\text{s.t. } B_{t+1} = RB_t + Y_t - C_t \quad (1')$$

$$B_t \geq -\bar{B}, \quad (2')$$

where $\bar{B}$ is a sufficiently high borrowing limit. The latter constraint is called the **no-Ponzi game condition** (NPGC) and does not allow the agent to choose an infinitely high consumption by borrowing more and more each period. Often, this restriction does not appear in the optimization problem and is only implicitly assumed. Write down the Lagrangian assuming that constraint (2') does not bind

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t u(C_t) + \sum_{t=0}^{\infty} \lambda_t (RB_t + Y_t - C_t - B_{t+1}).$$

Taking the optimality conditions, it is straightforward to show that the Euler equations (3) remain unchanged and hold for any period t . Remarkably, the transversality condition is also similar to the one above:

$$\lim_{T \rightarrow \infty} \beta^T u'(C_T) B_{T+1} = 0. \quad (4')$$

Following the same steps as above, derive the intertemporal budget constraint

$$\sum_{j=0}^{\infty} R^{-j} C_{t+j} = RB_t + \sum_{j=0}^{\infty} R^{-j} Y_{t+j}$$

and solve for the optimal consumption under the CRRA utility:

$$C_t = \left(1 - \beta^{\frac{1}{\sigma}} R^{\frac{1}{\sigma}-1}\right) \left(RB_t + \sum_{j=0}^{\infty} R^{-j} Y_{t+j}\right).$$

More generally, however, it is often not possible to solve the problem in closed form. For example, a model with stochastic shocks — a case that goes beyond the scope of this course, but will be discussed in detail in macro courses — leads to the following Euler equation

$$C_t^{-\sigma} = \beta R \mathbb{E}_t C_{t+1}^{-\sigma}.$$

It is not possible to express C_{t+1} from this condition and to substitute it into the budget constraint. Neither a finite horizon nor a continuous-time limit help to solve this problem. Instead, macroeconomists usually employ one of the following approaches (or their combination):

1. approximate solution based on the linearized equilibrium conditions,
2. recursive formulation of the problem and dynamic programming.

The next two lectures discuss in detail these two methods.

Continuous time Similarly to the infinite horizon, assuming that time is continuous rather than discrete can substantially simplify the analysis. This is especially true for the optimal stopping time problems, in which agents choose when to change price, buy/sell an asset, invest capital, etc. We start again with the baseline saving model, heuristically derive the continuous-time version of the problem and the optimality conditions.

The basic idea is to rewrite all equations assuming an arbitrary length of period Δ (instead of normalizing it to one) and then to take the limit of $\Delta \rightarrow 0$. To do this correctly, recall that macroeconomic variables can be divided into (i) *stocks* measured **at** a given point in time (e.g. capital, debt, prices) and (ii) *flows* measured **over** a period of time (e.g. inflation, interest, consumption, GDP). It follows that the former variables are independent of Δ , while the latter ones are proportional to the length of period. Applying this principle to the household budget constraint, we get

$$B_{t+\Delta} = (1 + r\Delta)B_t + Y_t\Delta - C_t\Delta,$$

where $r = R - 1$. Rewrite the equation in terms of growth rates

$$\frac{B_{t+\Delta} - B_t}{\Delta} = rB_t + Y_t - C_t$$

and take the continuous-time limit $\Delta \rightarrow 0$:

$$\dot{B}_t = rB_t + Y_t - C_t.$$

The left hand side of the equation $\dot{B}_t \equiv \lim_{\Delta \rightarrow 0} \frac{B_{t+\Delta} - B_t}{\Delta}$ corresponds the instantaneous change in wealth B_t in period t . To express the objective function in continuous time, it is convenient to define the discount factor $\rho > 0$ such that $\beta \equiv e^{-\rho\Delta}$. The discounted utility is then equal to

$$\sum_{t=0}^{\infty} e^{-\rho t} u(C_t) \Delta \longrightarrow \int_0^{\infty} e^{-\rho t} u(C_t) dt.$$

Hamiltonian Consider next a generalized **optimal control** problem:

$$\begin{aligned} \max_{\{C_t, B_t\}} \int_0^{\infty} e^{-\rho t} u(C_t) dt \\ \text{s.t. } \dot{B}_t = g(B_t, C_t), \end{aligned}$$

where C_t is a flow variable and B_t is a stock variable. Define the **Hamiltonian** as

$$\mathcal{H}_t \equiv e^{-\rho t} u(C_t) + \lambda_t g(B_t, C_t),$$

where λ_t is called a **co-state** variable. The Lagrangian of the discrete-time version of the problem can then be written as

$$\mathcal{L} = \sum_{t=0}^{\infty} \left[\mathcal{H}_t \Delta - \lambda_t (B_{t+\Delta} - B_t) \right].$$

Taking the first-order conditions and the limit $\Delta \rightarrow 0$, we get

$$\begin{aligned} \frac{\partial \mathcal{H}_t}{\partial C_t} \Delta = 0 & \Rightarrow \frac{\partial \mathcal{H}_t}{\partial C_t} = 0 \\ \frac{\partial \mathcal{H}_{t+\Delta}}{\partial B_{t+\Delta}} \Delta - \lambda_t + \lambda_{t+\Delta} = 0 & \Rightarrow \frac{\partial \mathcal{H}_t}{\partial B_t} = -\dot{\lambda}_t \end{aligned}$$

The intuition for these conditions is virtually the same as in the discrete time. The agent is free to sell and buy period- t goods at price λ_t in units of utility. The first optimality condition says that for each period (market) t , the household chooses consumption to maximize total utility. The second optimality condition is the no-arbitrage condition, which says that the price of a

period- t good is equalized with its returns $\lambda_t = \lambda_{t+\Delta} + \frac{\partial \mathcal{H}_t}{\partial B_t} \Delta$. As before, the transversality condition states

$$\lim_{t \rightarrow \infty} \lambda_t B_t = 0.$$

The **maximum principle** due to Pontryagin says that these three equations are the necessary conditions for optimality.

Example Consider again the savings problem with CRRA preferences and assume constant income $Y_t = Y$. The household problem in continuous time is

$$\begin{aligned} \max_{\{C_t, B_t\}_{t=0}^{\infty}} \int_0^{\infty} e^{-\rho t} \frac{C_t^{1-\sigma}}{1-\sigma} dt \\ \text{s.t. } \dot{B}_t = rB_t + Y - C_t. \end{aligned}$$

The Hamiltonian is given by

$$\mathcal{H}_t \equiv e^{-\rho t} \frac{C_t^{1-\sigma}}{1-\sigma} + \lambda_t (rB_t + Y - C_t)$$

and therefore, the optimality conditions are

$$\begin{aligned} 0 &= \frac{\partial \mathcal{H}_t}{\partial C_t} = e^{-\rho t} C_t^{-\sigma} - \lambda_t \\ -\dot{\lambda}_t &= \frac{\partial \mathcal{H}_t}{\partial B_t} = r\lambda_t \end{aligned}$$

Combine these equation to get the Euler equation:

$$\frac{\dot{C}_t}{C_t} = \frac{1}{\sigma}(r - \rho).$$

Solve this differential equation:

$$C_t = C_0 e^{\frac{r-\rho}{\sigma} t},$$

where C_0 is the (unknown) level of consumption in the initial period. This constant is determined by the intertemporal budget constraint. To obtain the latter, multiply both sides of the (flow) budget constraint by e^{-rt} and express it as

$$e^{-rt} \dot{B}_t - r e^{-rt} B_t = (Y - C_t) e^{-rt},$$

or equivalently as

$$d(e^{-rt} B_t) = (Y - C_t) e^{-rt} dt.$$

Integrate both sides from the initial period to period t :

$$e^{-rt}B_t = B_0 + \int_0^t (Y - C_s)e^{-rs}ds.$$

Next, use the transversality condition

$$\lim_{t \rightarrow \infty} \lambda_t B_t = 0,$$

which combined with the expression for λ_t and C_t implies $e^{-rt}B_t \rightarrow 0$ as $t \rightarrow \infty$. Applying this limit to the integrated budget constraint, obtain

$$B_0 + \int_0^\infty (Y - C_s)e^{-rs}ds = 0.$$

Finally, substitute the process for consumption

$$B_0 + \int_0^\infty (Y - C_0 e^{\frac{r-\rho}{\sigma}s})e^{-rs}ds = 0$$

and assuming r is not too high, solve

$$C_0 = \frac{\rho - r(1 - \sigma)}{\sigma} \left(B_0 + \frac{1}{r}Y \right).$$

Notice that the same expression can be obtained from the solution of the discrete-time model taking the limit $\Delta \rightarrow 0$.